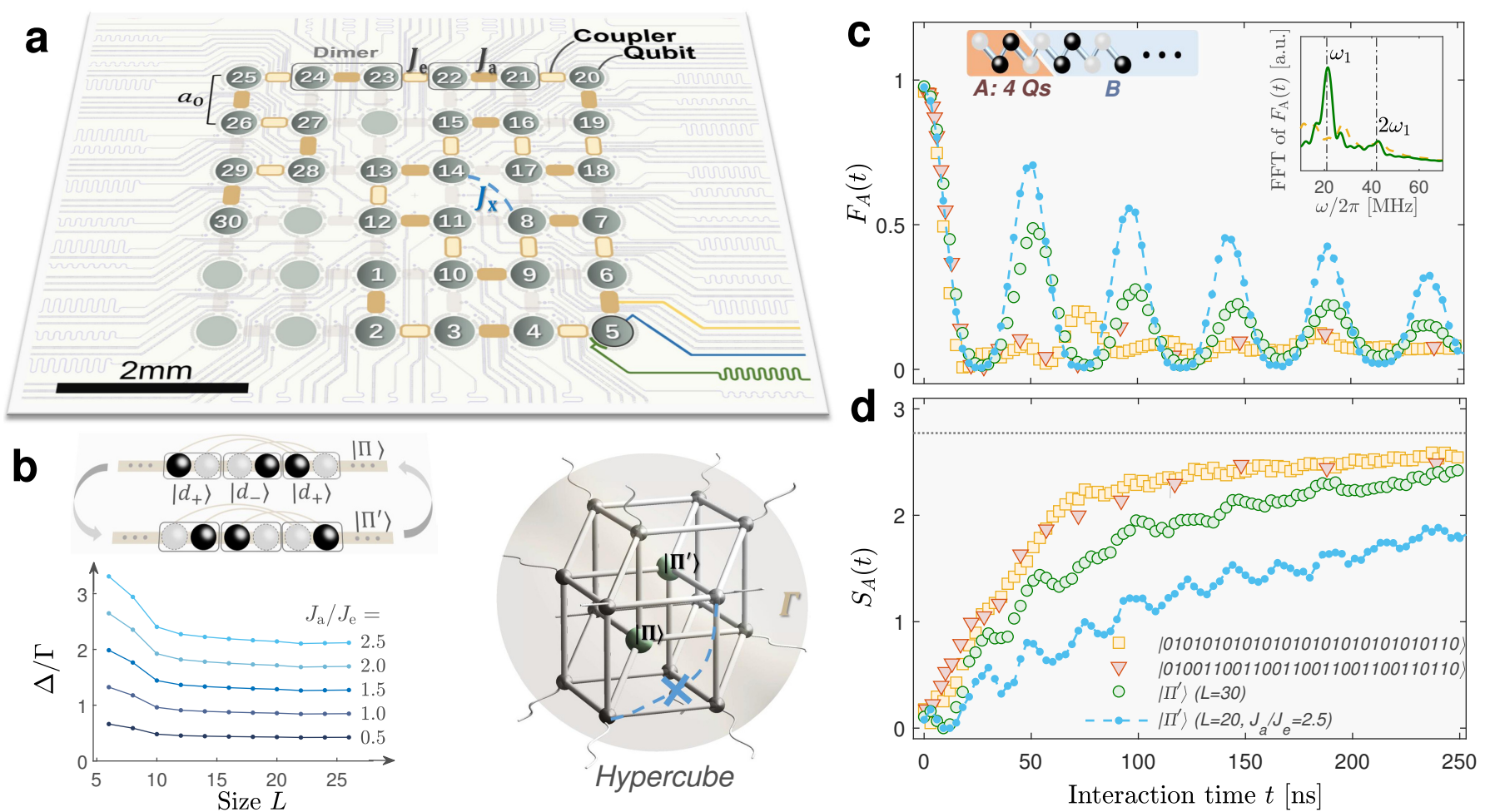

a
Dimer
J_e
J_a
Coupler
Qubit
a_o
25 24 23 22 21 20
26 27 15 16 19
29 28 13 14 17 18
J_x
30 12 11 8 7
1 10 9 6
2 3 4 5
2mm
b
$|\Pi\rangle$
$|d_+\rangle$ $|d_-\rangle$ $|d_+\rangle$
$|\Pi'\rangle$
Δ/Γ
J_a/J_e =
2.5
2.0
1.5
1.0
0.5
Size L
$|\Pi'\rangle$
$|\Pi\rangle$
Γ
Hypercube
c
A: 4 Qs
B
$F_A(t)$
FFT of $F_A(t)$ [a.u.]
ω_1
$2\omega_1$
$\omega/2\pi$ [MHz]
d
$S_A(t)$
$|01010101010101010101010101110\rangle$
$|01001100110011001100110011110\rangle$
$|\Pi'\rangle$ (L=30)
$|\Pi'\rangle$ (L=20, J_a/J_e=2.5)
Interaction time t [ns]